EP471 Homework 6
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Problem 1: Radioactive Decay of Bismuth-212

Assumptions made:

- The radioactive decay process is completely random and independent, perfectly modeled by a continuous exponential probability distribution.
- When evaluating the standard error of the mean for Monte Carlo estimates, the sample size (N) is assumed to be sufficiently large for the Central Limit Theorem to apply, meaning the sample mean is normally distributed.

(a) Derivation of the Cumulative Distribution Function (CDF) and Half-Life.

The probability density function (PDF) is given by $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$. The CDF, F(t), represents the probability that a decay occurs before or at time t and is found by integrating the PDF from 0 to t:

$$F(t) = \int_0^t \lambda e^{-\lambda \tau} d\tau$$

$$F(t) = \left[-e^{-\lambda \tau}\right]_0^t = -e^{-\lambda t} - (-e^0)$$

$$F(t) = 1 - e^{-\lambda t}$$

The half-life ($t_{1/2}$) is the time at which half of the material has decayed, meaning $F(t_{1/2}) = 0.5$. Substituting this into the CDF:

$$0.5 = 1 - e^{-\lambda t_{1/2}}$$

$$e^{-\lambda t_{1/2}} = 0.5$$

Taking the natural logarithm of both sides:

$$-\lambda t_{1/2} = \ln(0.5) = -\ln(2)$$

$$t_{1/2} = \frac{\ln(2)}{\lambda}$$

This mathematically confirms the relationship provided in the problem statement ($\lambda = \frac{\ln(2)}{t_{1/2}}$).

(b) Derivation of Expected Lifetime (μ_T) and Standard Deviation (σ_T):

The expected lifetime is the mean of the exponential distribution, defined as $\mu_T = E[T] = \int_0^\infty t f(t) dt$:

$$\mu_T = \int_0^\infty t \lambda e^{-\lambda t} dt$$

Using integration by parts, let $u = t$ and $dv = \lambda e^{-\lambda t} dt$. This gives $du = dt$ and $v = -e^{-\lambda t}$.

$$\mu_T = \left[-te^{-\lambda t}\right]_0^\infty - \int_0^\infty (-e^{-\lambda t}) dt$$

The first term evaluates to 0 because the exponential decay term $e^{-\lambda t}$ dominates the linear term t as $t \to \infty$. Evaluating the remaining integral:

$$\mu_T = \int_0^\infty e^{-\lambda t} dt = \left[-\frac{1}{\lambda} e^{-\lambda t} \right]_0^\infty = (0) - \left(-\frac{1}{\lambda} \right)$$

$$\mu_T = \frac{1}{\lambda}$$

To find the standard deviation, I first need the variance, $Var(T) = E[T^2] - (E[T])^2$.

$$E[T^2] = \int_0^\infty t^2 \lambda e^{-\lambda t} dt$$

Using integration by parts again (let $u = t^2$, $dv = \lambda e^{-\lambda t} dt$), I get:

$$E[T^2] = \left[-t^2 e^{-\lambda t} \right]_0^\infty + \int_0^\infty 2te^{-\lambda t} dt$$

$$E[T^2] = 0 + \frac{2}{\lambda} \int_0^\infty t\lambda e^{-\lambda t} dt$$

Notice the remaining integral is exactly the expected value $E[T]$ derived above:

$$E[T^2] = \frac{2}{\lambda} \left(\frac{1}{\lambda} \right) = \frac{2}{\lambda^2}$$

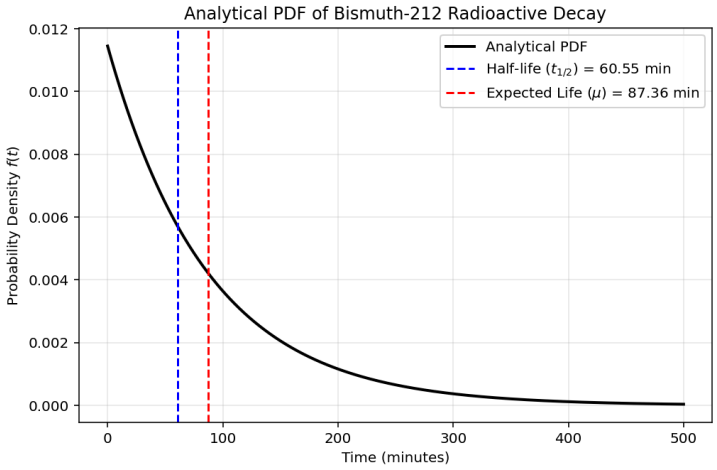
Now substitute this back into the variance formula:

$$Var(T) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 = \frac{1}{\lambda^2}$$

The standard deviation is the square root of the variance:

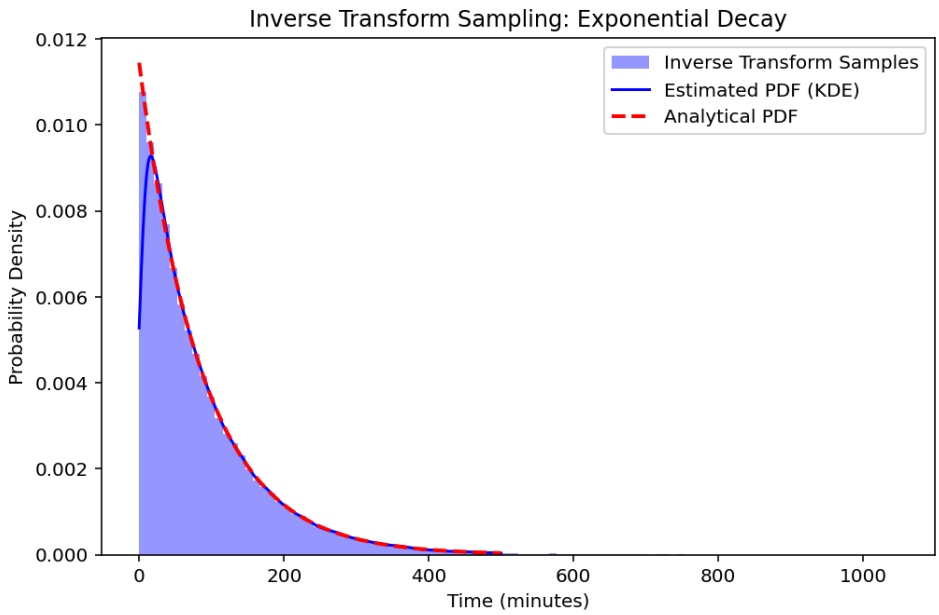
$$\sigma_T = \sqrt{\frac{1}{\lambda^2}} = \frac{1}{\lambda}$$

Plotting the analytical solution shows a confirmation of the concepts of half-life. The expected life is higher due to the mathematical derivation shown earlier, since decay is an exponential process.




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Analytical Results:
Decay constant (lambda): 0.0114 min^-1
Expected Lifetime (Mean): 87.36 min
Standard Deviation: 87.36 min
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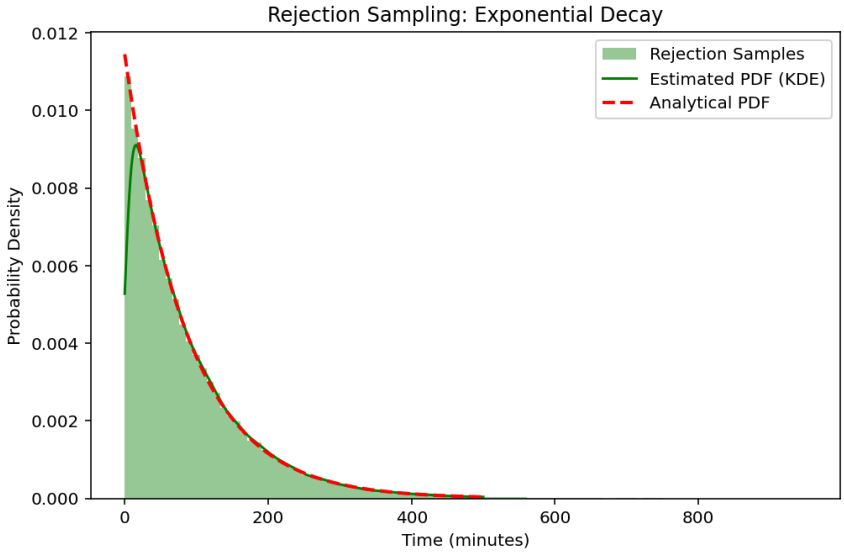
- (d) The Monte Carlo method is appropriate for estimation in this case because it mirrors decay's seemingly random nature while sticking to the trend that I assign it. Plotting the results from the inverse sampling method shows a close approximation to the analytical solution.




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Inverse Transform Sampling:
Estimated Mean: 87.00 min
Estimated Std Dev: 86.98 min
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The main error is at t=0, where an issue with KDE is exposed. The estimate at the boundary is often about 50% loIr than the true value because the "left half" of the probability kernels at the origin has nowhere to go. In reality, the probability of decay is highest at t=0.

- (e) Rejection sampling is a more complicated Monte Carlo method. Still, since I know that the maximum probability must be at t=0 and it cannot exceed lambda, I can easily set the limit for values.




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Rejection Sampling:
Estimated Mean: 87.75 min
Estimated Std Dev: 87.46 min
Total Candidates Generated: 1200000
Acceptance Rate: 0.0833
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The acceptance rate is extremely low with this method. This is likely due to the geometry of the PDF. The area in which samples are counted is in a box with a height of lambda and a width of 500. The analytical PDF takes up a small fraction of this box, leading to a low acceptance rate.

- (f) The standard error of the mean is $SE = \frac{\sigma_T}{\sqrt{N}}$. The relative error is the standard error divided by the true mean:

$$\text{Relative Error} = \frac{SE}{\mu_T} = \frac{\frac{\sigma_T}{\sqrt{N}}}{\mu_T}$$

Since I derived that $\mu_T = \sigma_T = \frac{1}{\lambda}$ for an exponential distribution, they cancel out:

$$\text{Relative Error} = \frac{1}{\sqrt{N}}$$

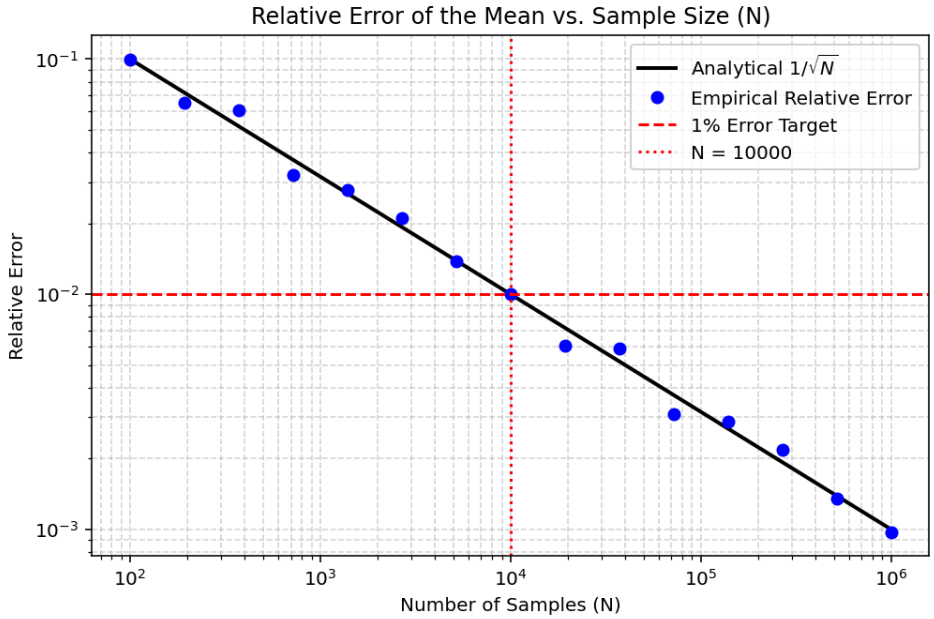
To reduce the relative error to below 1% (0.01):

$$\frac{1}{\sqrt{N}} \leq 0.01$$

$$\sqrt{N} \geq 100$$

$$N \geq 10,000$$

Thus, at least 10,000 samples are required.



Inverse transform sampling with different N values verifies the result of 10,000.

Problem 2: Probability-of-Failure for a Cantilever Beam

Assumptions & Hypotheses:

- The beam behaves as a linear elastic material, validating the deflection equation $\Delta = \frac{4QL^3}{Ebh^3}$
- All six input parameters (L, b, h, E, Q, Δ_L) are statistically independent random variables.

- Because I am focusing strictly on the deflection threshold, the limit state only considers deflection exceeding the allowable limit Δ_L, ignoring the stress-yielding failure mode mentioned in the prompt text for the sake of the mathematical formulation.

Derivations:

(a) Limit State Function Formulation In risk assessment, the limit state function $g(X)$ defines the boundary between safety and failure. A common formulation is

$g(X) = \text{Capacity} - \text{Demand}$ Failure occurs when $g(X) \leq 0$.

Here, the "Capacity" is the allowable deflection Δ_L, and the "Demand" is the actual deflection Δ. Therefore, the limit state function is:

$g(X) = \Delta_L - \Delta$

Substituting the physical response equation for the beam:

$$g(X) = \Delta_L - \frac{4QL^3}{Ebh^3}$$

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- Initial Monte Carlo Simulation:
Number of samples (N): 1000000
Number of failures: 30080
Estimated Probability of Failure (P_f): 0.030080
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- (c&d) The accuracy of a Monte Carlo simulation depends on the number of samples N. The coefficient of variation ($c.o.v.$) of the estimated probability of failure $\bar{P}$ represents the relative error and is given by:

$$c.o.v.(\bar{P}) = \sqrt{\frac{1 - P_f}{NP_f}}$$

To find the required sample size N for a desired target error ($c.o.v.$), I rearranged the equation:

$$(c.o.v.)^2 = \frac{1 - P_f}{NP_f}$$

$$N = \frac{1 - P_f}{P_f \times (c.o.v.)^2}$$

Using the P_f computed from the 1,000,000-sample simulation in part (b), I plug that P_f and a $c.o.v.$ of 0.01 (1%) into the equation above.

For Part (d): I am given a hypothetical $P_f \approx 10^{-10}$ and want an error of 2% ($c.o.v. = 0.02$).

$$N = \frac{1 - 10^{-10}}{10^{-10} \times (0.02)^2}$$

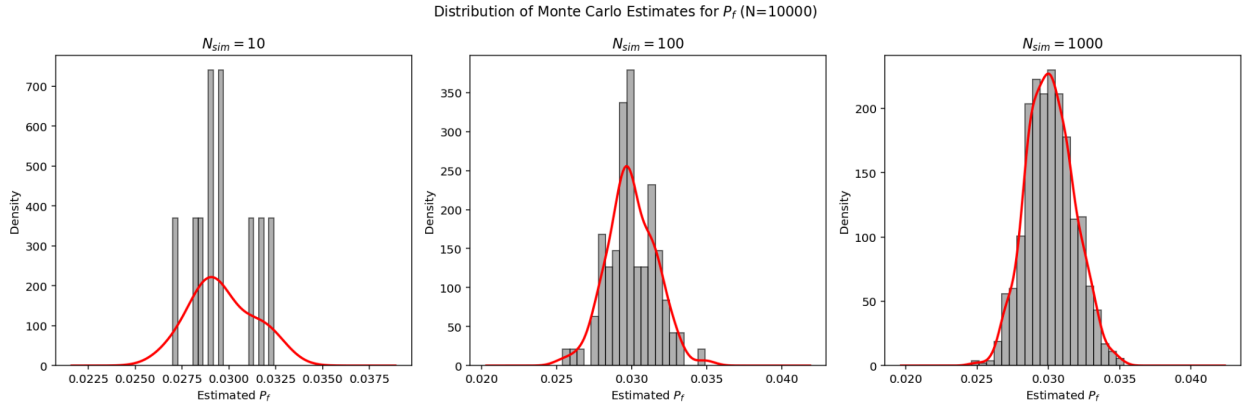
Since $1 - 10^{-10} \approx 1$, I can simplify:

$$N \approx \frac{1}{10^{-10} \times 0.0004} = \frac{1}{4 \times 10^{-14}}$$

$N \approx 2.5 \times 10^{13}$ samples

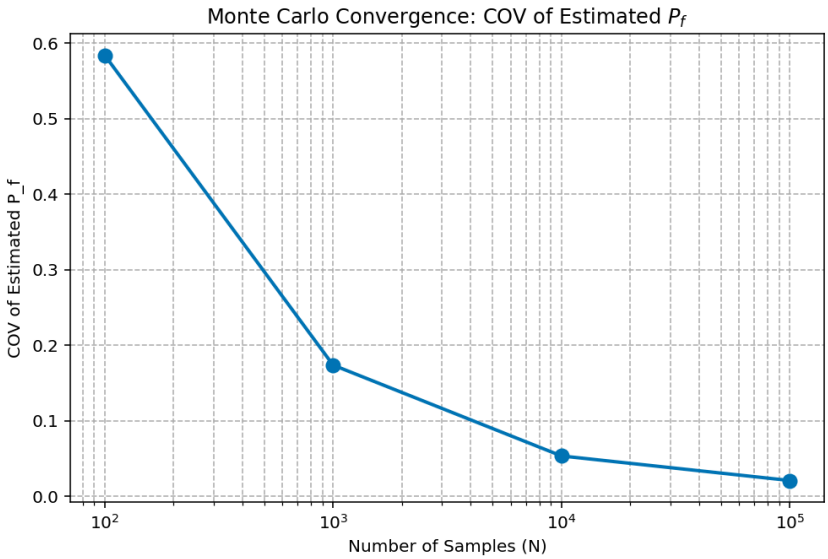
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- Sample Size Requirements:
Required samples for 1% error (based on estimated P_f = 0.030080): 322,447
Required samples for 2% error if P_f = 1e-10: 24,999,999,997,500
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- (e)



As N_{sim} increases, the sampling error (the uncertainty in the mean of the P_f estimates) decreases proportionally to $1/\sqrt{N_{sim}}$. This leads to a more stable and reliable estimate of the true probability of failure. While the variance of an individual run is determined by the sample size N, increasing the number of runs (N_{sim}) provides a more precise characterization of the estimator's behavior and ensures the results are not outliers. The distribution of P_f estimates increasingly resembles a Normal (Gaussian) distribution as N_{sim} grows. This is a direct consequence of the Central Limit Theorem, which states that the distribution of the average of a large number of independent, identically distributed variables will be approximately normal, regardless of the underlying distribution. In the plots, this is visible as the histogram becomes smoother and more symmetric, aligning more closely with the bell-shaped KDE curve.

- (f)



The plot illustrates the inverse relationship between the number of samples and the statistical uncertainty of the simulation:

- Monte Carlo Convergence: The trend follows the theoretical convergence rate of $1/\sqrt{N}$. As the sample size increases by a factor of 10, the Coefficient of Variation (COV) decreases by a factor of approximately 3.16 ($\sqrt{10}$).
- Precision and Stability: A higher N leads to a significant reduction in the COV, meaning the estimate of the probability of failure (P_f) becomes much more precise and reproducible.
- Diminishing Returns: Because of the square-root relationship, achieving extremely low error levels requires exponentially more computational power